[Title]

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Abstract

Electron backscatter diffraction (EBSD) orientation maps contain two kinds of error: a small scatter of every measured orientation, and misindexed pixels whose orientation is completely wrong. The filters commonly used to clean these maps work grain by grain, so the grains must first be detected with a threshold angle, and most of the filters have parameters that must be matched to the noise level of the map. Isolated misindexed pixels are not repaired by the filters themselves and have to be removed in a separate cleaning step. We train a convolutional neural network that cleans an orientation map without grain detection and without being told its noise level. For every pixel the network decides how much each of its neighbours should count, after aligning the neighbours across the 24 symmetry-equivalent forms of a cubic orientation, and a separate output detects and rebuilds misindexed pixels. The network was trained on about 39,000 synthetic DREAM.3D maps with randomly drawn noise levels and tested on 500 other maps under 28 noise conditions, with maximum scatter angles from 1° to 15° and 0 to 9% misindexed pixels. In every condition it removed 86–93% of the mean per-pixel error. On test maps with mixed noise levels the error fell from 5.69° to 0.59°, and 94% of the misindexed pixels ended within 5° of their true orientation. The standard deviation across three independent training runs was at most 0.03°. On 10 maps of every test set, the network had a lower mean error than the best of six MTEX filters in all 29 sets, significantly in 28 (paired t-test, Holm-corrected), and left misindexed pixels 2.0° to 3.1° from their true orientation against 5.6° to 8.0°. The network cleans a map in about 0.04 s on one GPU, including reading and writing the files, against 0.5 to 16 s per map for each MTEX filter on a desktop processor. Because the method needs only the orientation map, it can be applied to existing EBSD data, without the stored diffraction patterns that pattern-based denoising requires.

1 Introduction

[EBSD maps carry two kinds of error, scatter and misindexing. Current fixes are hand-tuned filters evaluated on about ten maps; deep-learning work cleans the diffraction patterns, which most laboratories do not store. Our contribution.]

2 Related work

[Map filters (Hielscher 2019, Atindama 2023); pattern denoising (Andrews 2023, Mani Krishna 2023, Prajapat 2026); deep learning for indexing and segmentation (Shen 2019, Ding 2020, Kaufmann 2021, Martinez Ostormujof 2022).]

3 Methods

3.1 Synthetic orientation maps

We generated 40,000 two-dimensional polycrystalline microstructures with DREAM.3D [1], each a 128×128 pixel orientation map on a square grid with a step of 1 μm and a single cubic phase (nickel, point group $m\bar{3}m$) [grain-size distribution and texture settings used in DREAM.3D]. A map contains on average 50 grains (standard deviation 4), the mean equivalent circle diameter of a grain is about 20 pixels, and 17% of the pixels lie on a grain boundary, which we define as a pixel with at least one of its four neighbours in a different grain. Two maps were incomplete because their generation run was interrupted and were discarded, leaving 39,998. Since every map is a separate microstructure, we split the data by map: maps 1–500 form the test set, maps 501–1000 the validation set, and the remaining 38,998 maps the training set. No grain therefore appears in more than one set. All maps were produced with the same generator settings, and the number of grains per map is the same across the three ranges.

3.2 Noise model

We model two kinds of error, each with its own level, so that a noisy map can contain one of them or both. The first is orientation scatter: every pixel is rotated by a small rotation about an axis drawn uniformly on the sphere, with a rotation angle drawn uniformly between 0 and $\theta_{\max}$. The disorientation between a scattered pixel and its true orientation therefore has a mean of $\theta_{\max}/2$. The second is misindexing: a fraction p of the pixels is replaced by orientations drawn uniformly from SO(3) [2], so that these pixels carry no information about their true orientation. In real scans, misindexing happens mostly at grain boundaries: there the electron beam covers two grains at once, the recorded diffraction pattern is a mixture of two patterns, and the indexing software may re-

turn an orientation that belongs to neither grain. To reproduce this, a boundary pixel is chosen four times as often as an interior pixel. No pixel is chosen twice, so exactly a fraction p of the pixels is replaced. Scatter is applied first and misindexing second. All errors are measured as the disorientation angle, the smallest rotation between two orientations once the 24 proper rotations of the cubic group are taken into account.

The network is trained on noise that is drawn afresh every time a training map is used. For each draw, $\theta_{\max}$ is sampled uniformly from $[0°, 15°]$ and p from $[0, 10\%]$. These are continuous ranges, not the fixed levels of the test sets (Section 3.3), and they contain every test level. The network therefore never sees the same corruption twice and is never told the noise level of its input. We also apply a random rotation of the specimen frame to all orientations of the map, which leaves every misorientation between pixels unchanged, together with random flips and 90° rotations of the pixel grid. The 500 validation maps were corrupted once, with a level drawn per map from the same ranges and a fixed random seed, so that every epoch and every training run is compared on identical inputs. The validation set is used only to select the checkpoint.

3.3 Test conditions

The test maps were corrupted under 28 fixed conditions: every combination of $\theta_{\max} \in \{1, 3, 5, 7, 9, 12, 15\}°$ and $p \in \{0, 3, 6, 9\}\%$, each applied to all 500 test maps (Fig. 1). The lowest level gives a mean error of 0.5°, comparable to the angular precision usually reported for Hough-based indexing [3]; at $\theta_{\max} = 15°$ (mean 7.5°) the grains are only just distinguishable. All 28 conditions were generated with the same random seed. For a given map, the rotation axes and the fraction of $\theta_{\max}$ drawn for each pixel are thus the same in every condition, and the scatter is identical across the four misindexing levels, so any difference between two conditions comes from the amount of noise and not from a different random draw. At $\theta_{\max} \geq 12°$, scatter alone moves 17–33% of the pixels beyond 10° from their true orientation. We therefore assess the repair of misindexed pixels on the recorded set of replaced pixels rather than with a fixed angular threshold.

3.4 Baseline filters

We compare our method with the six orientation-map filters of MTEX [4, 5], version 7.1.0. Each filter replaces the orientation of a pixel by one computed from the pixels around it; they differ in how these pixels are combined.

(i) *Mean filter.* The pixel is replaced by the average orientation of the 3×3 window centred on it, that is, of itself and its eight neighbours. It is the simplest and fastest filter, but it blurs any sharp change of orientation inside the window.

(ii) *Median filter.* Of the 25 orientations in the 5×5 window around the pixel, the one with the smallest mean disorientation to all the others replaces the pixel. The result is always one of the measured orientations, so a few outliers in the window cannot pull it away.

(iii) *Kuwahara filter* [6]. The 5×5 window is divided into four overlapping 3×3 quadrants, each with the pixel in one corner. The pixel takes the mean orientation of the quadrant in which the orientations are most alike. Near a sharp change this is the quadrant on the pixel's own side, so the change is not smeared.

(iv) *Spline filter* [7]. A smooth orientation field is fitted to the data, balancing closeness to the measured orientations against how strongly the field bends. The amount of smoothing is chosen automatically by generalised cross-validation, and a robust weighting reduces the influence of outliers. This is the default filter of MTEX.

(v) *Half-quadratic filter* [8]. The output map is the one that best balances closeness to the data against the differences between neighbouring pixels. As in total-variation denoising, both are measured with an ℓ_1-type norm, which favours uniform regions separated by sharp steps over a gradual blur. Differences larger than 15° are not penalised at all. We set the weight of the smoothing term to $\alpha = 0.25$, a quarter of the MTEX default.

(vi) *Infimal-convolution filter* [9]. Like (v), but the penalty combines first-order differences (weight $\lambda = 0.005$), which keep steps sharp, with second-order differences ($\mu = 0.01$), which allow the orientation to change gradually instead of breaking into a staircase of uniform patches. We use the MTEX default parameters.

MTEX applies a filter to one grain at a time, so that orientations are never averaged across a grain boundary. The grains are therefore first reconstructed from the noisy map: two neighbouring pixels belong to the same grain when their disorientation is below a threshold angle ω. Since each pixel can be up to $\theta_{\max}$ away from its true orientation, two pixels of the same grain can differ by up to $2\theta_{\max}$, and a smaller threshold breaks grains into fragments. We therefore set $\omega = \max(10°, 2\theta_{\max})$, using the true $\theta_{\max}$ of each test condition, or of each map in the mixed set. The filters are thus given the noise level, which our network never receives.

The filters alone cannot repair misindexed pixels. A misindexed pixel has a random orientation that almost never lies within ω of its neighbours, so it forms a grain of its own, and MTEX leaves grains of fewer than five

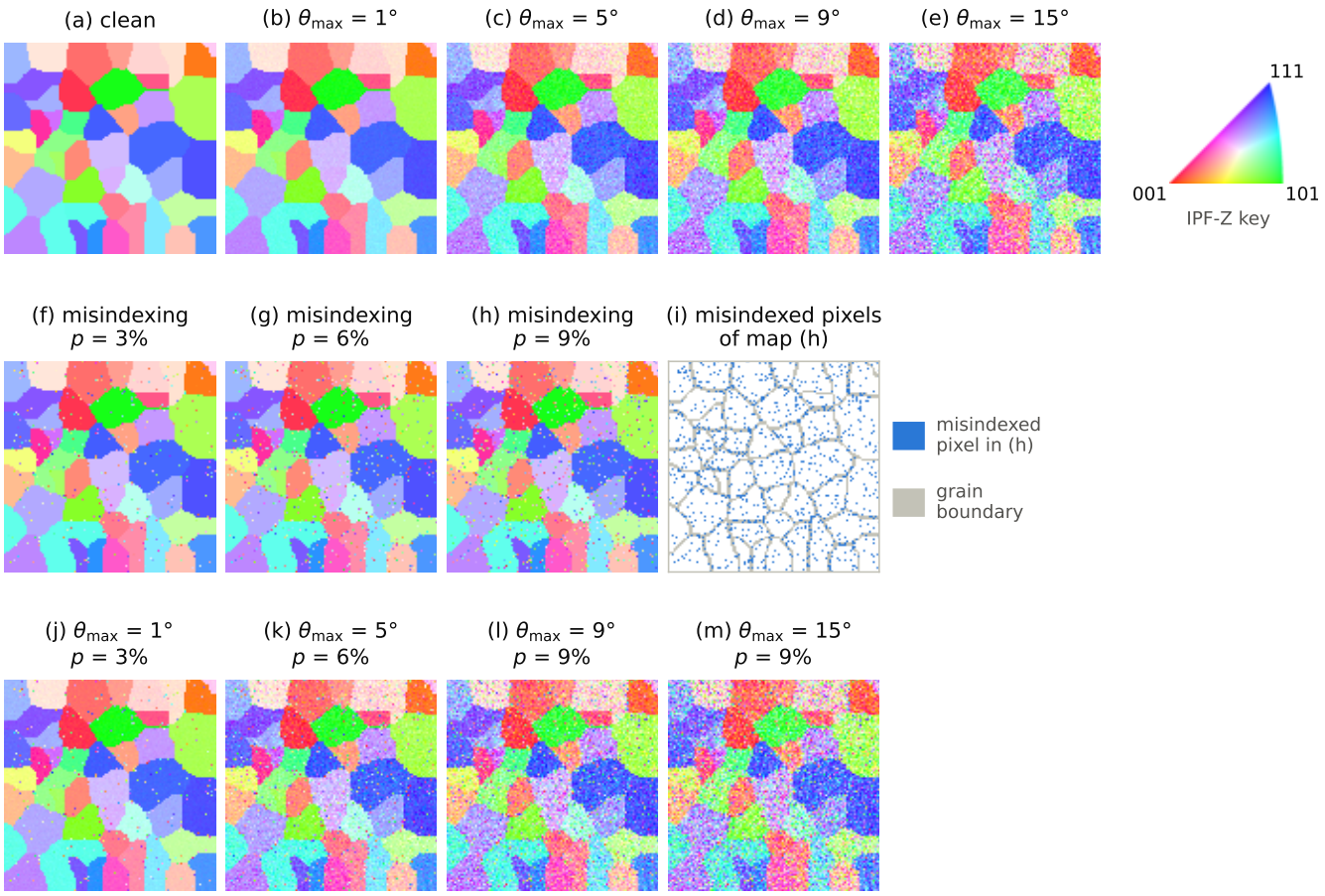


Figure 1: The noise model on test map 1, coloured with the IPF-Z key. (a) Clean map. (b)–(e) Orientation scatter alone, with maximum angle $\theta_{\max} = 1°$, 5°, 9° and 15°. (f)–(h) Misindexing alone, $p = 3$, 6 and 9% of the pixels: the replaced pixels of the corresponding test sets shown on the clean map, without their scatter. (i) Positions of the misindexed pixels in (h) (blue) relative to the grain boundaries (grey): boundary pixels make up 16% of the map but receive 41% of the misindexing, because a boundary pixel is four times as likely to be chosen. (j)–(m) Scatter and misindexing together, from the mildest to the most severe test condition: $\theta_{\max} = 1°$ with $p = 3\%$, 5° with 6%, 9° with 9%, and 15° with 9%.

pixels unfiltered. We therefore clean the map before filtering: grains smaller than three pixels are deleted, the grains are reconstructed without them, and each deleted pixel takes the orientation of the nearest remaining pixel, including at the edge of the map, so that it joins that pixel's grain. The filter is then applied to the cleaned map, grain by grain. We use the same procedure on every map. In the seven conditions without misindexing ($p = 0$) no pixel forms a grain of its own, so the cleaning removes nothing and the result is that of the filter alone.

The test maps were loaded into MTEX as .ang files without any change of reference frame, which we checked reproduces every orientation exactly. The filtered maps were written back on the original 128 × 128 grid and scored in the same way as the output of our network (Eq. 3). The filters were run on test maps 1–10 of each of the 28 test conditions and of the mixed set, 290 maps in total. On [processor], the six filters together took about 24 s per map, of which 16 s for the infimal-convolution filter, against about 0.04 s per map for our network on one NVIDIA H100 GPU, including reading and writing the files (500 maps in 21–23 s). The filter parameters were not tuned: they are the MTEX defaults, except for the window of the median and Kuwahara filters and α of the half-quadratic filter, [how these values were chosen].

3.5 Error measures

Disorientation. To compare a denoised map with the clean map we need the angle between two orientations. For a cubic crystal the same orientation can be written in 24 equivalent ways, so a plain difference between two sets of Euler angles is not a valid measure [10, 11]. We therefore use the disorientation angle. Let $\mathbf{q}$ and $\mathbf{q}'$ be

the unit quaternions of two orientations. Their misorientation is

$$\Delta\mathbf{q} = \mathbf{q}'\,\mathbf{q}^{-1}, \tag{1}$$

and the disorientation angle is the smallest rotation angle among the symmetrically equivalent forms of $\Delta\mathbf{q}$,

$$\delta(\mathbf{q},\mathbf{q}') = \min_{\mathbf{s}\in\mathcal{S}} \; 2\arccos\left|(\mathbf{s}\,\Delta\mathbf{q})_0\right|, \tag{2}$$

where $\mathcal{S}$ is the set of the 24 proper rotations of the cubic point group and $(\cdot)_0$ is the scalar part of a quaternion. Every error in this paper is a disorientation angle.

Four measures. A single number cannot show everything a denoising method does to a map, so we use four measures, each answering one question.

ALL: how large is the error overall? For a map m with the set of pixels Ω ($|\Omega| = 128 \times 128 = 16{,}384$), denoised map u_m and true map u_m^0, the error is the mean per-pixel disorientation,

$$e_m = \frac{1}{|\Omega|}\sum_{\mathbf{x}\in\Omega} \delta\big(u_m(\mathbf{x}), u_m^0(\mathbf{x})\big). \tag{3}$$

We use the mean, as Atindama et al. [10] did, and not the median. A median ignores any group of pixels smaller than half the map, so it would not change at all if the misindexed pixels, at most 9% of a test map, were left unrepaired.

BOUNDARY: are grain boundaries kept sharp? The same mean over the grain-boundary pixels B_m, the pixels with at least one of their four neighbours in a different grain of the true map. Since every DREAM.3D grain has a single orientation, these pixels are found exactly, without a threshold angle. About 17% of the pixels lie on a boundary. This is where a filter averages two grains and blurs the boundary, and where most misindexing occurs. A method can lower ALL by smoothing the grain interiors, which hold 83% of the pixels, while blurring the boundaries; BOUNDARY shows this.

CORRUPTED: were the misindexed pixels repaired? The same mean over the misindexed pixels M_m, which the noise generator replaced and recorded for every map. A replaced pixel starts on average 40.7° from its true orientation, so this measure shows directly whether a method found and rebuilt it. It is defined only for test sets with misindexing, and only for synthetic data, where the replaced pixels are known.

>10°: how many pixels are still grossly wrong? The percentage of pixels whose error exceeds 10°,

$$g_m = \frac{100}{|\Omega|}\sum_{\mathbf{x}\in\Omega} \mathbb{1}\Big[\delta\big(u_m(\mathbf{x}), u_m^0(\mathbf{x})\big) > 10^\circ\Big], \tag{4}$$

where $\mathbb{1}[\cdot]$ is 1 when the condition holds and 0 otherwise. A few wrong pixels barely move the mean: if 1% of the pixels are 60° off and the rest 0.3°, ALL is only 0.9°, yet about 160 pixels of the map are wrong. We chose 10° because it is a common threshold for a grain boundary in EBSD analysis, and the smallest grain threshold we give the MTEX filters (Section 3.4): a pixel more than 10° off would be taken for a separate grain, and so creates false grains and boundaries. A smaller threshold, such as 5°, would count ordinary scatter as gross error: at $\theta_{\max} = 9^\circ$, scatter alone moves 44% of the pixels beyond 5° but none beyond 10°. Conversely, only 0.67% of random orientation pairs lie within 10° of each other, so practically every misindexed pixel that is not repaired is counted. At $\theta_{\max} \geq 12^\circ$, scatter alone moves 17–33% of the pixels beyond 10° (Section 3.3). At these levels the measure mixes the two kinds of error, and CORRUPTED is the measure of misindexing repair.

Averaging over maps. Each test set has $N = 500$ maps. We compute every measure per map first and then report the mean over the maps, for example $\bar{e} = \frac{1}{N}\sum_{m=1}^{N} e_m$, with the standard deviation across the maps. The map, not the pixel, is the unit of every statistic: neighbouring pixels share a grain and a filter window, so their errors are not independent, and treating the 16,384 pixels of a map as independent samples would overstate the certainty of a result. Since every map has the same number of pixels, $\bar{e}$ is also the mean disorientation over all $N|\Omega|$ pixels of the set.

Repeated training. We trained the network three times, with different random seeds. Unless stated otherwise, the value given for our method is the mean of the three runs, and the ± value is the standard deviation across the runs, which shows how much a result depends on the random start of training. A difference between two methods smaller than this spread is not a result.

Fraction of noise removed. The input error grows with $\theta_{\max}$, so absolute errors cannot be compared between noise levels. We therefore also give the fraction of the input error that the denoising removed,

$$R = 1 - \frac{\bar{e}_{\text{out}}}{\bar{e}_{\text{in}}}, \tag{5}$$

which puts all levels on the same scale.

Statistical test. To test whether two methods differ, we use the paired t-test on the per-map errors [12], as Atindama et al. [10] did. The test is paired because both methods are applied to the same maps with the same noise, so taking the difference map by map removes the variation between easy and hard maps. For the same n maps let $d_m = e_m^{(1)} - e_m^{(2)}$ be the difference between the errors of the two methods on map m, $\bar{d}$ the mean and S_d the standard deviation of the d_m. The test statistic

$$T = \frac{\bar{d}}{S_d/\sqrt{n}} \tag{6}$$

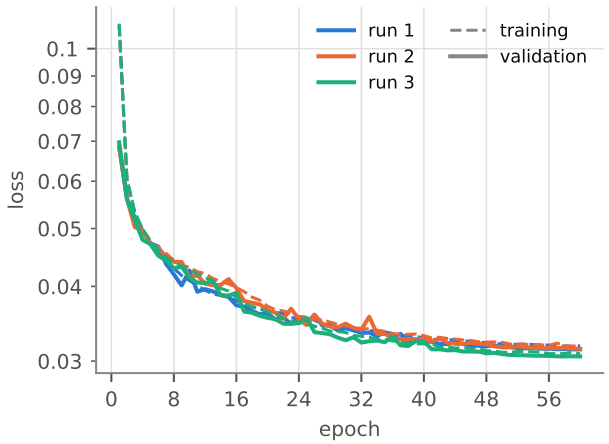


Figure 2: Training loss (dashed) and validation loss (solid) of the three training runs. The two losses stay close throughout, which shows that the network does not overfit the training maps.

has a t-distribution with $n-1$ degrees of freedom. We take $p < 0.05$ as statistically significant.

4 Results

4.1 Training

Figure 2 shows the training and validation loss of the three runs. Each run had 60 epochs of 2,437 batches of 16 maps, about 5.8 min per epoch and 5.8 h per run on one NVIDIA H100 GPU. The training and validation losses fall together and stay close at every epoch, so the network does not overfit the training maps. Both flatten after about epoch 45. For each run we kept the checkpoint with the best validation score, which came from epochs 58, 50 and 58. On the validation maps the gate, the part of the network that marks a pixel as misindexed, found 98.8% of the misindexed pixels (recall), and 99.9% of the pixels it marked were truly misindexed (precision). The three runs gave the same values to the third decimal.

4.2 Maps with mixed noise levels

We first report one overall result. In the mixed test set, each of the 500 test maps has its own noise level, drawn at random from the same ranges as in training: $\theta_{\max}$ between 0° and 15° and p between 0 and 10%. Table 1 shows the four error measures. The mean per-pixel error drops from 5.69° to 0.589 ± 0.015°, so 89.6% of the error is removed. The misindexed pixels, which are 40.7° from their true orientation on average, end 2.47° from it, and the percentage of pixels more than 10° wrong falls from 11.27% to 0.23%.

Table 1: Mean per-pixel errors on the mixed test set (500 maps, each with its own random noise level). For our method, mean ± standard deviation over three training runs. R is the fraction of the error removed, Eq. (5).

Measure	Noisy input	Ours	R
ALL (°)	5.692	0.589 ± 0.015	89.6%
BOUNDARY (°)	8.554	1.197 ± 0.020	86.0%
CORRUPTED (°)	40.709	2.466 ± 0.032	93.9%
$> 10°$ (% pixels)	11.27	0.226 ± 0.002	98.0%

4.3 Effect of the noise level

Tables 2 and 3 give the results for the 28 test conditions, and Figure 3 plots the errors against the maximum scatter angle. The standard deviation across the three runs is at most 0.032° for the mean error and 0.006 percentage points for the percentage of gross errors, far smaller than any of the differences discussed below.

The error left after denoising grows in proportion to the scatter. With scatter alone ($p = 0$), the mean error goes from 0.068° at $\theta_{\max} = 1°$ to 0.922° at $\theta_{\max} = 15°$, but the fraction removed stays between 86.5% and 88.1% at every level. This is what we expect from averaging. The network replaces each pixel by a weighted average of its neighbours in the same grain, and averaging random errors shrinks them by a fixed factor, whatever their size. So the remaining error is a fixed fraction of the input error. When misindexing is added, the fraction removed is larger, between 88.1% and 93.4%, because the misindexed pixels carry a large part of the input error and most of them are repaired.

With misindexing, the percentage of grossly wrong pixels (Table 3) falls from about p in the input to between 0.10% and 0.58% after denoising. At $\theta_{\max} = 12°$ and 15°, scatter alone puts 16.7% and 33.4% of the pixels beyond 10°; after denoising only 0.02% and 0.05% remain.

4.4 Repair of misindexed pixels

A misindexed pixel carries no information about its true orientation, so it must be rebuilt from its neighbours. Over the 21 test conditions with misindexing, between 91.7% ($\theta_{\max} = 15°$, $p = 9\%$) and 95.5% ($\theta_{\max} = 1°$, $p = 3\%$) of the misindexed pixels end within 5° of their true orientation. Their mean error drops from about 40.7° to between 2.02° and 3.26° (Fig. 3, right). The repair becomes slightly harder as the scatter grows. A likely reason is that a misindexed pixel whose random orientation happens to lie close to the true one is hard to tell apart from a strongly scattered pixel.

4.5 Where the remaining errors are

Figure 4 shows one test map under three noise conditions. The denoised maps are close to the true map of Fig. 1(a),

Table 2: Mean per-pixel error, ALL (°), for the 28 test conditions, 500 maps each: maximum scatter angle $\theta_{\max}$ and percentage of misindexed pixels p. Ours is the mean of three training runs (standard deviation across runs $\leq 0.032°$).

	$p = 0\%$		$p = 3\%$		$p = 6\%$		$p = 9\%$	
$\theta_{\max}$ (°)	Input	Ours	Input	Ours	Input	Ours	Input	Ours
1	0.500	0.068	1.708	0.128	2.913	0.196	4.121	0.273
3	1.500	0.186	2.679	0.248	3.853	0.317	5.031	0.396
5	2.501	0.301	3.649	0.365	4.793	0.437	5.942	0.518
7	3.501	0.418	4.619	0.485	5.733	0.560	6.852	0.645
9	4.501	0.538	5.589	0.609	6.673	0.688	7.762	0.777
12	6.002	0.724	7.044	0.803	8.084	0.891	9.127	0.989
15	7.502	0.922	8.500	1.013	9.494	1.111	10.492	1.222

Table 3: Percentage of pixels more than 10° from their true orientation, Eq. (4), for the same conditions as Table 2. Ours is the mean of three training runs (standard deviation across runs ≤ 0.006 percentage points).

	$p = 0\%$		$p = 3\%$		$p = 6\%$		$p = 9\%$	
$\theta_{\max}$ (°)	Input	Ours	Input	Ours	Input	Ours	Input	Ours
1	0.00	0.00	2.98	0.10	5.96	0.23	8.94	0.37
3	0.00	0.00	2.98	0.11	5.96	0.23	8.94	0.38
5	0.00	0.00	2.98	0.11	5.96	0.24	8.94	0.39
7	0.00	0.00	2.98	0.12	5.96	0.25	8.94	0.41
9	0.00	0.00	2.98	0.13	5.96	0.27	8.94	0.43
12	16.66	0.02	19.14	0.15	21.62	0.31	24.10	0.49
15	33.35	0.05	35.33	0.21	37.30	0.38	39.29	0.58

even at $\theta_{\max} = 15°$ where the grains in the input are hard to see. In the error maps of the denoised maps, the remaining error follows the grain boundaries. The numbers confirm this. In every test condition, between 92.5% and 99.7% of the pixels that remain more than 10° wrong lie on a grain boundary, although boundary pixels make up only 17% of a map. With misindexing, most of these pixels (72–98%) are misindexed pixels on a boundary. Such a pixel has neighbours from two grains, and the network sometimes gives it the orientation of the wrong grain. For the same reason the BOUNDARY error is always higher than the ALL error (Fig. 3, middle).

4.6 Comparison with MTEX filters

The six MTEX filters (Section 3.4) were run on test maps 1–10 of every test set, 290 maps in total, and our network was scored on the same maps, so all methods were given identical noisy pixels. For each set we compare the network with the best filter of that set, the one with the lowest ALL error. This filter is chosen after scoring, which favours MTEX.

Figure 5 shows the ALL error of every method in the 28 conditions of the test grid. The network has the lowest error in all of them and in the mixed set. The median and spline filters are the strongest MTEX filters and are the best filter of every set; the mean, Kuwahara, half-quadratic and infimal-convolution filters are well behind. The gap widens with the scatter: the best filter's error is 1.2 times ours at $\theta_{\max} = 1°$ without misindexing (0.081° against 0.068°) and 3.8 times at $\theta_{\max} = 15°$ (3.49° against 0.92°). At $\theta_{\max} = 15°$ the mean filter even leaves the map worse than the input, at every misindexing level (9.80° against 7.51° without misindexing).

Within each set we tested the difference with the paired t-test over the 10 maps, Holm-corrected over the 29 sets (Table 4; every set is in Appendix A). The network is significantly better than the best filter in 28 of the 29 sets for ALL and for BOUNDARY, in all 22 sets with misindexing for CORRUPTED, and in 25 sets for the percentage of pixels beyond 10°. It is significantly worse in none. The one set without a significant difference in ALL and BOUNDARY is $\theta_{\max} = 3°$ without misindexing, where the spline filter reaches 0.222° against our 0.185° ($p = 0.43$). Its error varies strongly from map to map (standard deviation 0.142°, against 0.005° for ours), and it is lower than ours on 5 of the 10 maps. Beyond 10°, the four sets without a significant difference are $\theta_{\max} = 1°$, 3° and 5° without misindexing, where both methods leave almost no pixel above 10° (at most 0.2 pixels per map on average), and the mixed set (0.27% against 1.94%, $p = 0.053$).

The largest difference is in the repair of misindexed pixels. After the best filter, a misindexed pixel is still

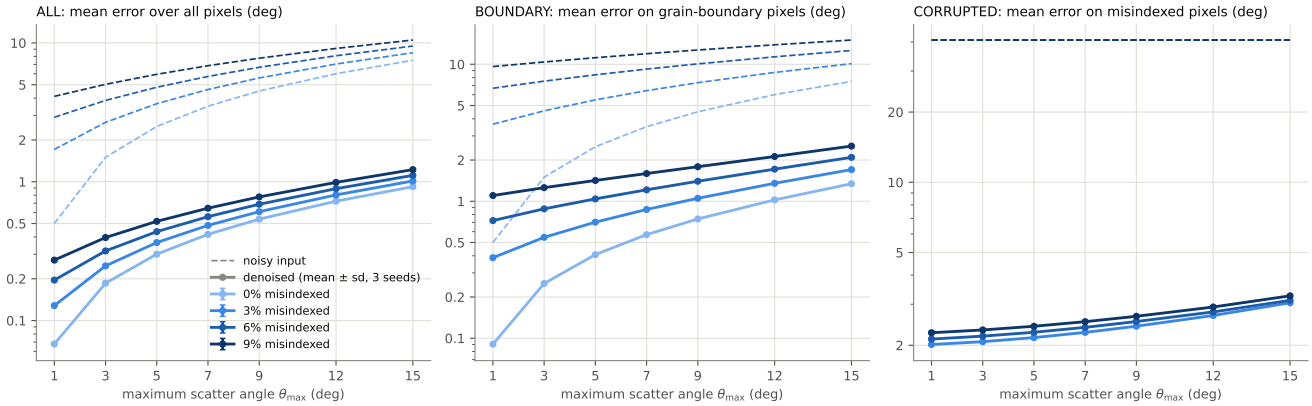


Figure 3: Mean per-pixel error against the maximum scatter angle, for each percentage of misindexed pixels: (left) all pixels, (middle) grain-boundary pixels, (right) misindexed pixels. Dashed lines: noisy input. Solid lines with markers: our method, mean ± standard deviation over three training runs (the error bars are smaller than the markers). Logarithmic vertical axes.

5.6° to 8.0° from its true orientation on average, depending on the condition, and after the network 2.0° to 3.1°. The filters fill a deleted pixel from its nearest neighbour and then smooth it, whereas the network rebuilds it from its whole neighbourhood. Figure 6 shows where the remaining errors lie. After the median filter they follow the grain boundaries (Fig. 6g), and at $\theta_{\max} = 15°$ whole grains are left more than 10° off (Fig. 6o). The network's error is lower everywhere and only faintly marks the boundaries (Fig. 6h, p).

Three limits of this comparison should be kept in mind. The filters were run with near-default settings rather than tuned for each condition (Section 3.4). They were, on the other hand, given the true scatter level through the grain threshold, which the network never receives. Finally, with 10 maps per set the per-set tests have little power, which matters only for the closest set, $\theta_{\max} = 3°$.

5 Discussion

[Why the error falls by a fixed fraction; what the network learned; limitations: synthetic training data, cubic symmetry only, 2D maps.]

6 Conclusion

[Conclusion]

Table 4: The network against the best MTEX filter of each test set, on test maps 1–10. Seven of the 29 sets are shown; all are in Appendix A. Each value is the mean (standard deviation) over the 10 maps; for the network, each map's value is the mean of the three training runs. The best MTEX filter is the one with the lowest ALL error on the set, chosen after scoring, and is used for all four measures. All MTEX filters were run with pre-cleaning. The lower of the two methods is in bold. Stars: paired t-test of the network against that filter over the 10 maps, Holm-corrected over the 29 sets: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; no star: not significant. Last column: difference in ALL, network minus filter, with its 95% confidence interval.

$\theta_{\max}$, p	Method	ALL (°)	BOUNDARY (°)	CORRUPTED (°)	>10° (%)	ΔALL [95% CI]
1°, 0%	Noisy input	0.500 (0.003)	0.496 (0.006)	–	0.00 (0.00)	
	Median (MTEX)	0.081 (0.002)	0.113 (0.006)	–	**0.00** (0.00)	
	Ours	**0.068** (0.002)***	**0.090** (0.004)***	–	**0.00** (0.00)	−0.014 [−0.015, −0.012]
5°, 0%	Noisy input	2.502 (0.016)	2.479 (0.032)	–	0.00 (0.00)	
	Median (MTEX)	0.408 (0.014)	0.580 (0.041)	–	0.01 (0.02)	
	Ours	**0.300** (0.009)***	**0.404** (0.014)***	–	**0.00** (0.00)	−0.108 [−0.114, −0.102]
15°, 0%	Noisy input	7.506 (0.048)	7.437 (0.096)	–	33.37 (0.41)	
	Median (MTEX)	3.485 (0.667)	9.436 (1.804)	–	7.66 (2.46)	
	Ours	**0.919** (0.025)***	**1.342** (0.046)***	–	**0.06** (0.02)***	−2.566 [−3.045, −2.086]
1°, 9%	Noisy input	4.112 (0.015)	9.695 (0.292)	40.613 (0.174)	8.94 (0.02)	
	Spline (MTEX)	0.628 (0.048)	3.395 (0.205)	5.644 (0.287)	1.34 (0.08)	
	Ours	**0.268** (0.009)***	**1.087** (0.072)***	**2.210** (0.108)***	**0.36** (0.03)***	−0.360 [−0.392, −0.327]
5°, 9%	Noisy input	5.933 (0.018)	11.220 (0.278)	40.613 (0.174)	8.94 (0.02)	
	Spline (MTEX)	0.940 (0.200)	3.712 (0.242)	5.875 (0.503)	1.61 (0.65)	
	Ours	**0.511** (0.012)***	**1.390** (0.071)***	**2.315** (0.123)***	**0.37** (0.03)**	−0.429 [−0.571, −0.288]
15°, 9%	Noisy input	10.488 (0.039)	15.033 (0.249)	40.613 (0.174)	39.33 (0.33)	
	Median (MTEX)	3.992 (0.390)	11.746 (1.152)	7.944 (0.535)	8.89 (1.35)	
	Ours	**1.211** (0.026)***	**2.477** (0.067)***	**3.117** (0.147)***	**0.54** (0.06)***	−2.781 [−3.060, −2.502]
mixed	Noisy input	6.806 (2.026)	10.089 (2.760)	40.681 (0.524)	14.52 (11.70)	
	Median (MTEX)	1.553 (0.860)	4.685 (2.487)	6.629 (0.589)	1.94 (1.85)	
	Ours	**0.724** (0.268)**	**1.476** (0.534)**	**2.584** (0.377)***	**0.27** (0.14)	−0.829 [−1.263, −0.396]

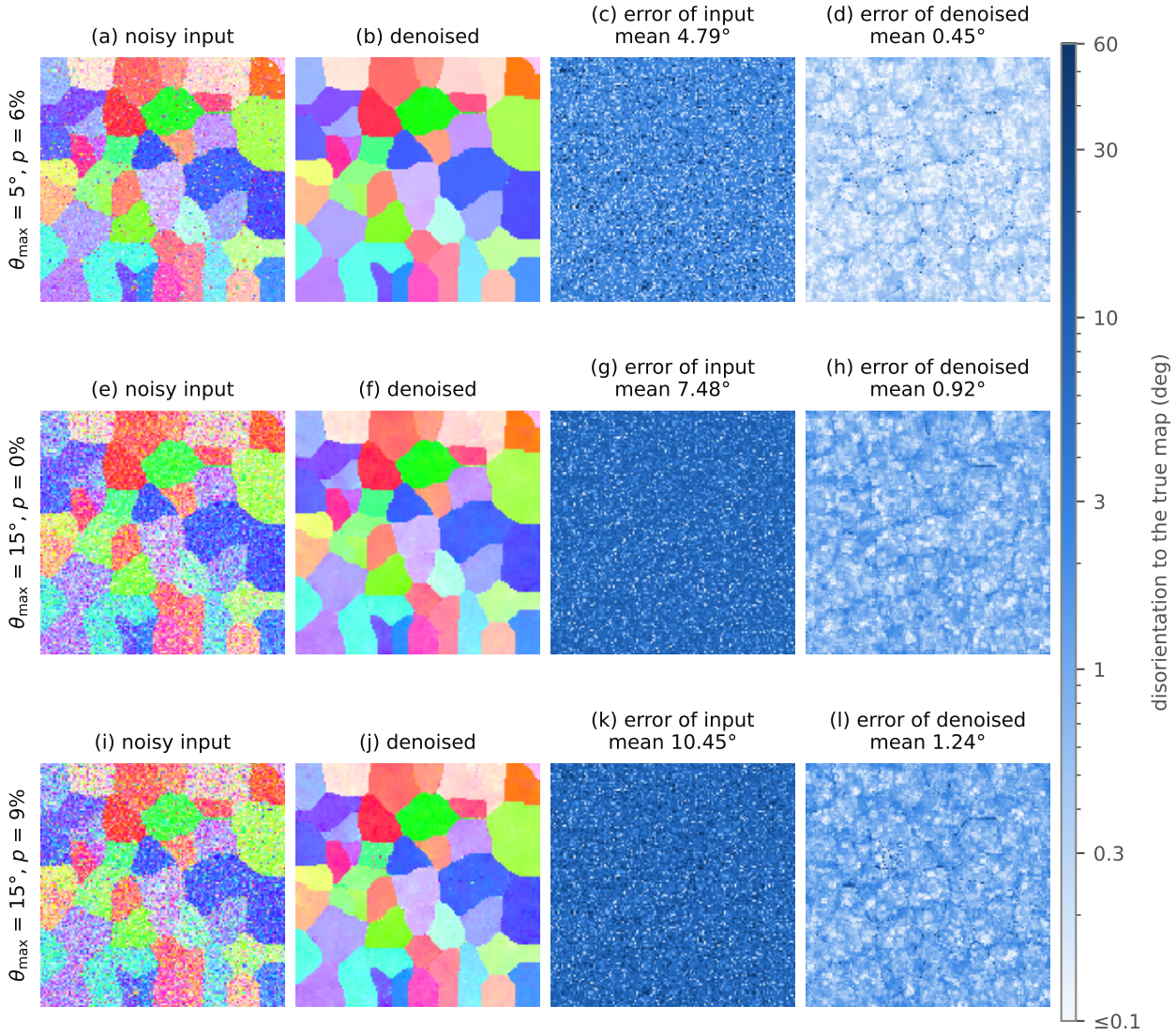


Figure 4: Test map 1 (Fig. 1(a)) denoised by our method (run 1) under three noise conditions, one per row. From left: noisy input and denoised map (IPF-Z colours), and the disorientation of each to the true map (logarithmic colour scale). The mean per-pixel error of each map is given above it. The remaining error follows the grain boundaries.

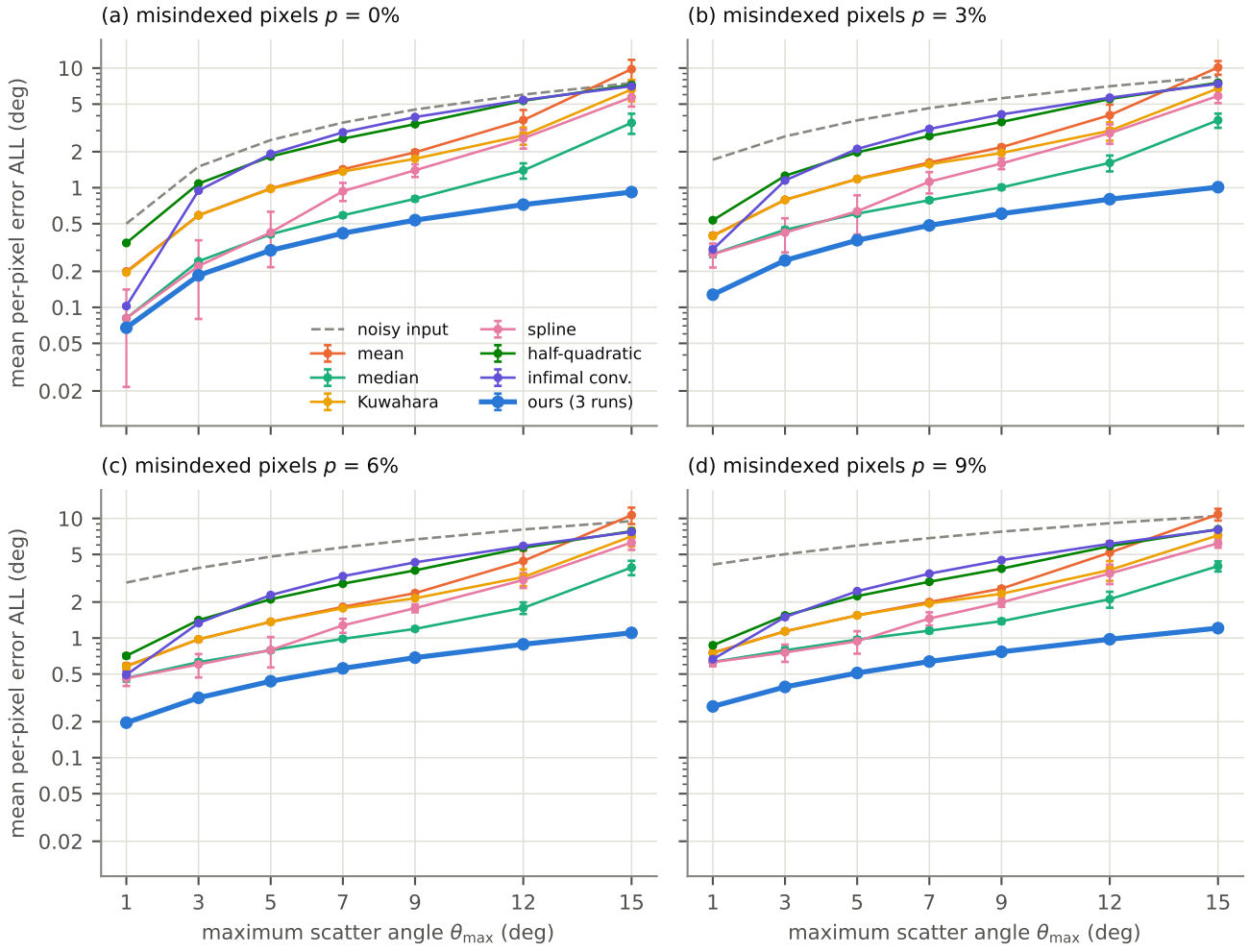


Figure 5: Mean per-pixel error ALL of the network and of the six MTEX filters (all with pre-cleaning) on test maps 1–10 of each condition, against the maximum scatter angle; one panel per misindexing level p. Points: mean over the 10 maps; bars: standard deviation across them (for the network, each map's value is the mean of the three training runs). Dashed: the noisy input. Note the logarithmic error axis.

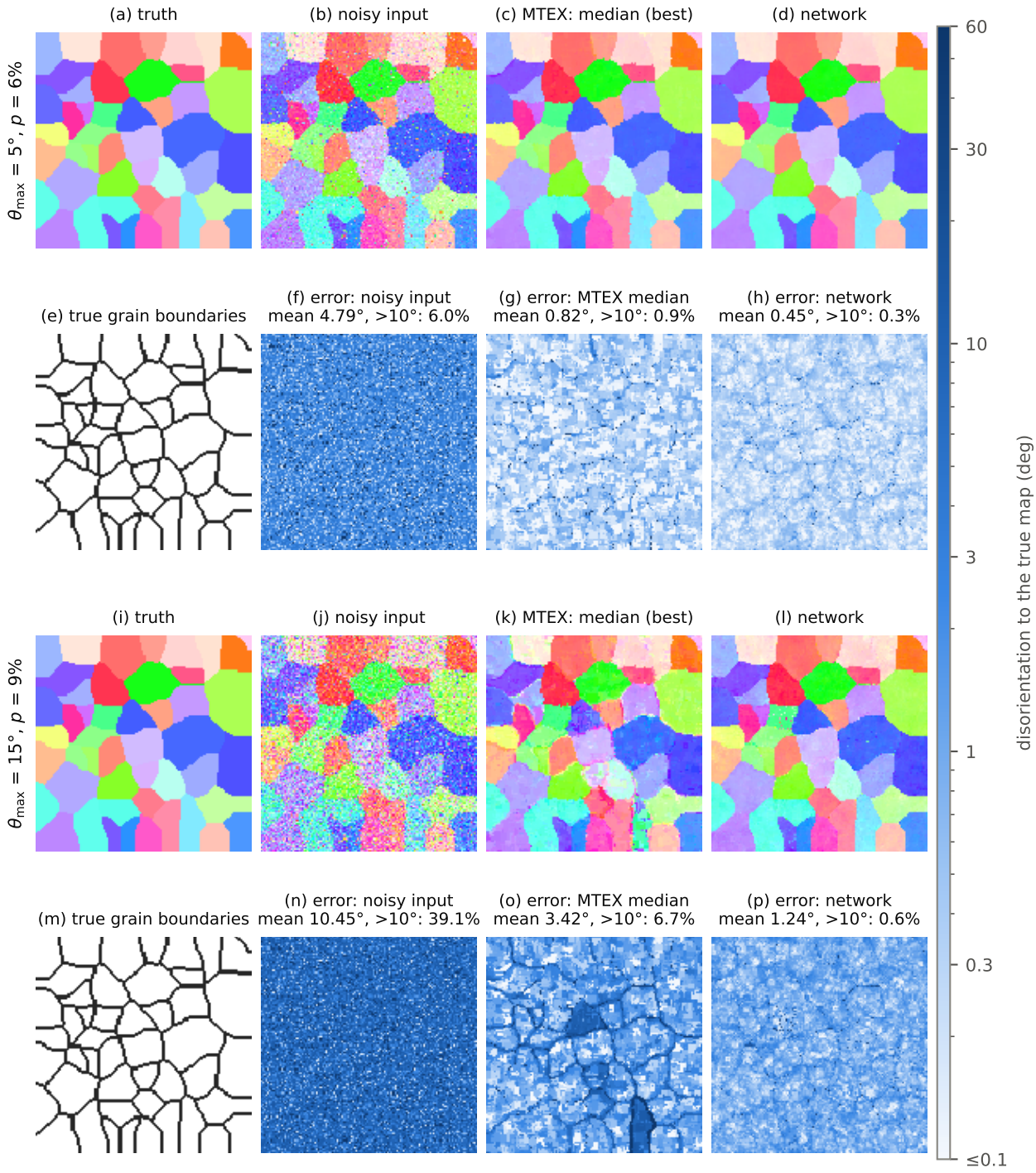


Figure 6: Test map 1 cleaned by the best MTEX filter of the set (the median filter in both cases) and by the network (training run 1), coloured with the IPF-Z key. (a)–(h) $\theta_{\max} = 5°$, $p = 6\%$; (i)–(p) $\theta_{\max} = 15°$, $p = 9\%$. Top row of each: the true map, the noisy input and the two cleaned maps. Bottom row: the true grain boundaries and the disorientation of each map to the truth, on one logarithmic colour scale, with its mean and the percentage of pixels beyond 10°.

A Full comparison with the MTEX filters

Tables 5–8 give every test set and every MTEX filter (all with pre-cleaning), on test maps 1–10. Each value is the mean over the 10 maps; the standard deviations are in `figures/mtex_comparison/summary.csv` and `per_map.csv`. The lowest value of each row, the noisy input aside, is in bold. Δ is the network minus the best filter of the set (the one with the lowest ALL error), with its 95% confidence interval, and p the paired t-test over the 10 maps, Holm-corrected over the sets.

Table 5: ALL: network against every MTEX filter, all test sets with their maps 1–10.

$\theta_{\max}$, p	Input	Mean	Median	Kuwah.	Spline	Half-q.	Inf. conv.	Ours	Δ [95% CI]	p
1°, 0%	0.500	0.199	0.081	0.196	0.081	0.345	0.102	**0.068**	−0.014 [−0.015, −0.012]	<0.001
3°, 0%	1.501	0.589	0.243	0.587	0.222	1.082	0.947	**0.185**	−0.037 [−0.138, +0.065]	0.43
5°, 0%	2.502	0.986	0.408	0.977	0.423	1.824	1.916	**0.300**	−0.108 [−0.114, −0.102]	<0.001
7°, 0%	3.503	1.427	0.588	1.365	0.935	2.567	2.902	**0.416**	−0.172 [−0.182, −0.161]	<0.001
9°, 0%	4.504	1.975	0.807	1.747	1.399	3.399	3.894	**0.536**	−0.272 [−0.288, −0.255]	<0.001
12°, 0%	6.005	3.674	1.395	2.735	2.587	5.330	5.407	**0.722**	−0.673 [−0.819, −0.526]	<0.001
15°, 0%	7.506	9.800	3.485	6.626	5.706	7.245	7.049	**0.919**	−2.566 [−3.045, −2.086]	<0.001
1°, 3%	1.714	0.398	0.279	0.394	0.278	0.535	0.304	**0.128**	−0.151 [−0.197, −0.104]	<0.001
3°, 3%	2.685	0.792	0.445	0.790	0.421	1.257	1.150	**0.247**	−0.175 [−0.272, −0.077]	0.006
5°, 3%	3.655	1.184	0.607	1.179	0.635	1.974	2.109	**0.364**	−0.243 [−0.250, −0.235]	<0.001
7°, 3%	4.626	1.624	0.786	1.571	1.122	2.708	3.097	**0.484**	−0.302 [−0.321, −0.284]	<0.001
9°, 3%	5.597	2.180	1.006	1.952	1.595	3.545	4.098	**0.607**	−0.400 [−0.423, −0.376]	<0.001
12°, 3%	7.053	4.043	1.613	2.994	2.856	5.517	5.656	**0.802**	−0.811 [−0.987, −0.635]	<0.001
15°, 3%	8.509	10.129	3.664	6.763	5.853	7.539	7.387	**1.011**	−2.653 [−3.016, −2.291]	<0.001
1°, 6%	2.910	0.584	0.463	0.579	0.461	0.710	0.493	**0.196**	−0.265 [−0.312, −0.219]	<0.001
3°, 6%	3.851	0.976	0.629	0.975	0.602	1.411	1.334	**0.317**	−0.285 [−0.382, −0.189]	<0.001
5°, 6%	4.792	1.368	0.791	1.365	0.794	2.110	2.288	**0.435**	−0.355 [−0.375, −0.335]	<0.001
7°, 6%	5.733	1.823	0.984	1.773	1.276	2.847	3.288	**0.558**	−0.426 [−0.450, −0.403]	<0.001
9°, 6%	6.673	2.375	1.193	2.152	1.778	3.678	4.292	**0.686**	−0.507 [−0.531, −0.483]	<0.001
12°, 6%	8.084	4.408	1.787	3.234	3.057	5.687	5.889	**0.887**	−0.900 [−1.042, −0.758]	<0.001
15°, 6%	9.496	10.642	3.887	7.080	6.291	7.819	7.711	**1.106**	−2.781 [−3.157, −2.405]	<0.001
1°, 9%	4.112	0.754	0.631	0.747	0.628	0.870	0.665	**0.268**	−0.360 [−0.392, −0.327]	<0.001
3°, 9%	5.023	1.140	0.790	1.137	0.757	1.543	1.497	**0.390**	−0.367 [−0.457, −0.276]	<0.001
5°, 9%	5.933	1.550	0.971	1.548	0.940	2.241	2.464	**0.511**	−0.429 [−0.571, −0.288]	<0.001
7°, 9%	6.844	2.000	1.153	1.945	1.456	2.955	3.452	**0.636**	−0.517 [−0.550, −0.483]	<0.001
9°, 9%	7.755	2.588	1.383	2.344	1.987	3.804	4.478	**0.767**	−0.615 [−0.653, −0.578]	<0.001
12°, 9%	9.122	5.180	2.114	3.705	3.466	5.870	6.137	**0.978**	−1.137 [−1.369, −0.905]	<0.001
15°, 9%	10.488	10.818	3.992	7.220	6.194	8.116	8.052	**1.211**	−2.781 [−3.060, −2.502]	<0.001
mixed	6.806	3.825	1.553	2.862	2.437	4.347	4.582	**0.724**	−0.829 [−1.263, −0.396]	0.006

Table 6: BOUNDARY: network against every MTEX filter, all test sets with their maps 1–10.

$\theta_{\max}$, p	Input	Mean	Median	Kuwah.	Spline	Half-q.	Inf. conv.	Ours	Δ [95% CI]	p
1°, 0%	0.496	0.226	0.113	0.245	0.107	0.451	0.170	**0.090**	−0.023 [−0.026, −0.019]	<0.001
3°, 0%	1.487	0.643	0.337	0.735	0.265	1.438	1.090	**0.249**	−0.016 [−0.146, +0.114]	0.79
5°, 0%	2.479	1.094	0.580	1.221	0.514	2.427	2.065	**0.404**	−0.176 [−0.198, −0.153]	<0.001
7°, 0%	3.470	1.728	0.900	1.697	1.342	3.418	3.048	**0.566**	−0.334 [−0.379, −0.288]	<0.001
9°, 0%	4.462	2.862	1.424	2.155	2.387	4.409	4.032	**0.738**	−0.686 [−0.788, −0.585]	<0.001
12°, 0%	5.949	6.511	3.183	3.500	5.394	5.907	5.531	**1.020**	−2.163 [−2.673, −1.653]	<0.001
15°, 0%	7.437	17.493	9.436	9.619	14.629	7.409	7.141	**1.342**	−8.094 [−9.395, −6.792]	<0.001
1°, 3%	3.731	1.412	1.300	1.429	1.292	1.618	1.367	**0.392**	−0.900 [−0.982, −0.817]	<0.001
3°, 3%	4.643	1.832	1.534	1.922	1.459	2.564	2.272	**0.549**	−0.910 [−1.009, −0.810]	<0.001
5°, 3%	5.554	2.234	1.738	2.372	1.689	3.463	3.180	**0.709**	−1.030 [−1.060, −0.999]	<0.001
7°, 3%	6.465	2.837	2.042	2.845	2.458	4.421	4.155	**0.875**	−1.168 [−1.259, −1.076]	<0.001
9°, 3%	7.376	3.955	2.550	3.278	3.506	5.411	5.148	**1.050**	−1.500 [−1.622, −1.377]	<0.001
12°, 3%	8.743	7.871	4.327	4.697	6.778	7.036	6.750	**1.354**	−2.973 [−3.568, −2.379]	<0.001
15°, 3%	10.110	18.914	10.617	10.764	15.941	8.781	8.567	**1.705**	−8.912 [−9.837, −7.988]	<0.001
1°, 6%	6.841	2.513	2.401	2.527	2.391	2.700	2.477	**0.739**	−1.652 [−1.783, −1.521]	<0.001
3°, 6%	7.675	2.907	2.617	2.993	2.532	3.572	3.336	**0.889**	−1.643 [−1.791, −1.495]	<0.001
5°, 6%	8.508	3.295	2.810	3.424	2.712	4.415	4.213	**1.047**	−1.763 [−1.904, −1.623]	<0.001
7°, 6%	9.342	3.952	3.182	3.971	3.526	5.410	5.234	**1.214**	−1.968 [−2.138, −1.798]	<0.001
9°, 6%	10.175	4.985	3.596	4.354	4.556	6.352	6.197	**1.400**	−2.196 [−2.310, −2.082]	<0.001
12°, 6%	11.425	8.893	5.189	5.671	7.782	8.035	7.842	**1.709**	−3.480 [−3.927, −3.032]	<0.001
15°, 6%	12.676	20.045	11.425	11.811	17.559	10.008	9.855	**2.079**	−9.345 [−10.081, −8.609]	<0.001
1°, 9%	9.695	3.522	3.409	3.530	3.395	3.686	3.489	**1.087**	−2.308 [−2.445, −2.170]	<0.001
3°, 9%	10.457	3.846	3.553	3.916	3.463	4.445	4.262	**1.236**	−2.227 [−2.328, −2.126]	<0.001
5°, 9%	11.220	4.318	3.844	4.440	3.712	5.340	5.211	**1.390**	−2.322 [−2.497, −2.148]	<0.001
7°, 9%	11.982	4.900	4.120	4.889	4.473	6.210	6.134	**1.557**	−2.563 [−2.753, −2.373]	<0.001
9°, 9%	12.745	6.051	4.636	5.355	5.629	7.237	7.192	**1.745**	−2.891 [−3.078, −2.704]	<0.001
12°, 9%	13.889	10.705	6.555	7.053	9.348	9.049	8.963	**2.072**	−4.483 [−5.144, −3.822]	<0.001
15°, 9%	15.033	20.008	11.746	12.470	17.297	11.292	11.189	**2.477**	−9.269 [−10.111, −8.427]	<0.001
mixed	10.089	7.692	4.685	5.278	6.529	6.722	6.519	**1.476**	−3.209 [−4.614, −1.805]	0.001

Table 7: CORRUPTED: network against every MTEX filter, all test sets with misindexing.

$\theta_{\max}$, p	Input	Mean	Median	Kuwah.	Spline	Half-q.	Inf. conv.	Ours	Δ [95% CI]	p
1°, 3%	40.919	6.490	6.379	6.492	6.369	6.516	6.476	**2.002**	−4.367 [−4.692, −4.043]	<0.001
3°, 3%	40.919	6.911	6.598	6.918	6.559	6.974	7.344	**2.054**	−4.505 [−4.854, −4.155]	<0.001
5°, 3%	40.919	7.079	6.571	7.111	6.551	7.163	7.966	**2.141**	−4.430 [−4.699, −4.162]	<0.001
7°, 3%	40.919	7.465	6.711	7.488	7.016	7.772	9.028	**2.263**	−4.448 [−4.868, −4.027]	<0.001
9°, 3%	40.919	7.937	6.867	7.819	7.505	8.822	10.364	**2.379**	−4.488 [−5.077, −3.898]	<0.001
12°, 3%	40.919	9.499	6.985	8.685	8.761	11.798	13.269	**2.650**	−4.336 [−4.609, −4.062]	<0.001
15°, 3%	40.919	14.873	8.021	11.279	11.753	17.041	17.958	**2.982**	−5.040 [−5.539, −4.540]	<0.001
1°, 6%	40.668	6.161	6.045	6.156	6.043	6.179	6.150	**2.139**	−3.904 [−4.295, −3.514]	<0.001
3°, 6%	40.668	6.494	6.177	6.498	6.133	6.535	6.909	**2.180**	−3.953 [−4.312, −3.594]	<0.001
5°, 6%	40.668	6.735	6.205	6.737	6.170	6.807	7.636	**2.256**	−3.949 [−4.215, −3.683]	<0.001
7°, 6%	40.668	7.329	6.567	7.356	6.825	7.622	8.870	**2.360**	−4.207 [−4.615, −3.799]	<0.001
9°, 6%	40.668	7.684	6.574	7.605	7.227	8.517	10.090	**2.496**	−4.078 [−4.284, −3.872]	<0.001
12°, 6%	40.668	9.251	6.597	8.312	8.409	11.437	12.913	**2.719**	−3.878 [−4.139, −3.616]	<0.001
15°, 6%	40.668	14.848	7.778	11.278	11.804	16.650	17.557	**3.024**	−4.754 [−5.397, −4.111]	<0.001
1°, 9%	40.613	5.768	5.649	5.760	5.644	5.775	5.753	**2.210**	−3.434 [−3.670, −3.198]	<0.001
3°, 9%	40.613	6.012	5.679	6.001	5.636	6.036	6.431	**2.250**	−3.386 [−3.706, −3.067]	<0.001
5°, 9%	40.613	6.477	5.936	6.472	5.875	6.524	7.371	**2.315**	−3.560 [−3.890, −3.230]	<0.001
7°, 9%	40.613	6.884	6.087	6.872	6.365	7.134	8.413	**2.414**	−3.673 [−4.030, −3.316]	<0.001
9°, 9%	40.613	7.477	6.344	7.380	7.024	8.240	9.814	**2.539**	−3.805 [−4.129, −3.481]	<0.001
12°, 9%	40.613	9.757	6.727	8.570	8.702	11.279	12.771	**2.777**	−3.950 [−4.228, −3.671]	<0.001
15°, 9%	40.613	14.938	7.944	11.456	11.544	16.699	17.660	**3.117**	−4.828 [−5.210, −4.445]	<0.001
mixed	40.681	8.931	6.629	8.131	7.964	9.815	11.030	**2.584**	−4.046 [−4.374, −3.717]	<0.001

Table 8: >10°: network against every MTEX filter, all test sets with their maps 1–10.

$\theta_{\max}$, p	Input	Mean	Median	Kuwah.	Spline	Half-q.	Inf. conv.	Ours	Δ [95% CI]	p
1°, 0%	0.00	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	+0.00 [−0.00, +0.00]	0.24
3°, 0%	0.00	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	+0.00 [−0.00, +0.00]	0.24
5°, 0%	0.00	**0.00**	0.01	**0.00**	0.30	**0.00**	**0.00**	**0.00**	−0.01 [−0.02, +0.00]	0.24
7°, 0%	0.00	0.04	0.04	**0.00**	0.22	**0.00**	**0.00**	**0.00**	−0.04 [−0.05, −0.02]	0.002
9°, 0%	0.00	0.45	0.12	0.02	0.41	**0.00**	**0.00**	**0.00**	−0.11 [−0.15, −0.08]	<0.001
12°, 0%	16.62	5.12	1.06	2.10	3.03	12.41	9.14	**0.02**	−1.04 [−1.48, −0.60]	0.002
15°, 0%	33.37	27.62	7.66	17.32	15.27	31.21	29.12	**0.06**	−7.60 [−9.37, −5.83]	<0.001
1°, 3%	2.98	0.48	0.48	0.48	0.48	0.48	0.48	**0.11**	−0.37 [−0.40, −0.35]	<0.001
3°, 3%	2.98	0.49	0.49	0.49	0.49	0.49	0.49	**0.11**	−0.38 [−0.40, −0.35]	<0.001
5°, 3%	2.98	0.47	0.48	0.48	0.88	0.48	0.48	**0.11**	−0.37 [−0.39, −0.35]	<0.001
7°, 3%	2.98	0.49	0.50	0.48	0.76	0.52	0.53	**0.12**	−0.38 [−0.42, −0.35]	<0.001
9°, 3%	2.98	0.91	0.56	0.48	0.90	0.60	0.62	**0.12**	−0.44 [−0.49, −0.38]	<0.001
12°, 3%	19.09	6.27	1.51	2.87	3.93	12.90	10.10	**0.16**	−1.35 [−1.84, −0.87]	0.001
15°, 3%	35.34	29.95	7.82	18.50	15.68	31.83	30.11	**0.21**	−7.61 [−8.99, −6.23]	<0.001
1°, 6%	5.96	0.93	0.93	0.93	0.93	0.93	0.93	**0.23**	−0.70 [−0.74, −0.65]	<0.001
3°, 6%	5.96	0.93	0.93	0.93	0.93	0.93	0.93	**0.23**	−0.70 [−0.75, −0.64]	<0.001
5°, 6%	5.96	0.91	0.92	0.92	1.31	0.92	0.93	**0.24**	−0.69 [−0.71, −0.66]	<0.001
7°, 6%	5.96	0.95	0.95	0.94	1.12	1.02	1.05	**0.24**	−0.71 [−0.76, −0.66]	<0.001
9°, 6%	5.96	1.32	0.97	0.93	1.32	1.21	1.23	**0.26**	−0.71 [−0.76, −0.66]	<0.001
12°, 6%	21.56	7.20	1.84	3.50	4.57	13.42	10.98	**0.29**	−1.55 [−1.95, −1.14]	<0.001
15°, 6%	37.33	31.92	8.51	19.80	17.47	32.45	31.15	**0.36**	−8.15 [−9.45, −6.85]	<0.001
1°, 9%	8.94	1.34	1.34	1.34	1.34	1.35	1.34	**0.36**	−0.98 [−1.05, −0.91]	<0.001
3°, 9%	8.94	1.32	1.32	1.32	1.32	1.32	1.32	**0.37**	−0.95 [−1.02, −0.89]	<0.001
5°, 9%	8.94	1.34	1.35	1.35	1.61	1.35	1.36	**0.37**	−1.24 [−1.69, −0.78]	0.001
7°, 9%	8.94	1.33	1.34	1.34	1.51	1.49	1.52	**0.38**	−0.96 [−1.03, −0.89]	<0.001
9°, 9%	8.94	1.78	1.39	1.35	1.81	1.80	1.83	**0.40**	−0.99 [−1.07, −0.92]	<0.001
12°, 9%	24.06	10.05	2.76	5.49	5.83	14.02	12.12	**0.45**	−2.30 [−2.96, −1.64]	<0.001
15°, 9%	39.33	32.46	8.89	20.06	16.87	33.08	32.19	**0.54**	−8.35 [−9.33, −7.37]	<0.001
mixed	14.52	7.20	1.94	4.19	4.11	8.59	7.60	**0.27**	−1.67 [−2.89, −0.44]	0.05

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